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Tuwaiq Academy

Agentic AI

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Capstone Project

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AGENTICAI CAPSTONE PROJECT

The Capstone Project serves as the culminating experience of the Agentic AI Bootcamp.

It is designed as a rigorous, large-scale engineering challenge that requires students to architect, implement, and evaluate a fully functional agentic AI system using advanced design patterns and production-grade methodologies.

Unlike traditional AI projects that focus solely on model training or prompt engineering, this capstone emphasizes the design and construction of intelligent, multi-component agentic workflows that demonstrate autonomy, memory management, tool use, planning capabilities, safety guardrails, and systematic evaluation.

Under instructor supervision, students will identify a real-world problem that requires more than simple question-answering. The selected problem must necessitate a structured agentic architecture incorporating at least several of the following design patterns:

- Reflection
- ReAct (Reason–Act–Observe)
- Planning
- Tool Use and Code Execution
- Memory Management (short-term and long-term)
- Guardrails and Human-in-the-Loop governance
- Evaluation and Error Analysis frameworks
- Reasoning Based Agents

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DETAILED COURSE LEARNING OBJECTIVES

Students will design a system that includes:

- A well-defined agent architecture
- Clear separation of components
- Structured task decomposition
- Explicit communication patterns (for multi-agent systems, if applicable)
- Proper memory integration
- Safety enforcement layers
- Measurable evaluation metrics

The project must go beyond prototype-level demos and demonstrate engineering depth. Students are expected to justify architectural decisions, evaluate trade-offs, and implement modular systems that are maintainable, scalable, and observable.

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DETAILED COURSE LEARNING OBJECTIVES

Each project must include a documented evaluation framework, including:

- End-to-end performance metrics
- Component-level evaluation
- Error analysis methodology
- Evidence of iterative improvement

Students are also required to address ethical considerations, system safety, and potential misuse scenarios. Guardrails must be implemented and justified within the architecture.

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DETAILED COURSE LEARNING OBJECTIVES

The final deliverable will include:

- **Working agentic system**
- **Technical architecture documentation with Code**
- **A Working Prototype**
- **Design pattern explanation**
- **Evaluation results**
- **Recorded demonstration**
- **Formal presentation**

This capstone is intentionally demanding. It evaluates not just technical implementation, but architectural thinking, engineering discipline, safety awareness, and the ability to build reliable agentic systems suitable for real-world deployment.

By the completion of this project, students should demonstrate the ability to engineer autonomous AI systems that are modular, safe, measurable, and scalable.

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PROBLEM DEFINITION & SOLUTION DESIGN (25%)

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
Problem Clarity & Relevance (10%)	Problem is exceptionally well-defined, highly significant, and clearly addresses a real-world need with strong justification.	Problem is clearly defined and relevant, addressing a discernible real-world need.	Problem is generally defined and has some relevance, but may lack depth or strong justification.	Problem definition is vague or relevance is unclear; justification is weak.	Problem is poorly defined, irrelevant, or lacks any clear real-world application.
Solution Design & Architecture (15%)	AI solution design is highly innovative, robust, scalable, and perfectly aligned with the problem. Architecture is well-justified and expertly detailed.	AI solution design is effective, appropriate, and well-structured. Architecture is clearly explained and mostly logical.	AI solution design is adequate and addresses the problem, but may have minor inefficiencies or less optimal architectural choices.	AI solution design is incomplete, conceptually flawed, or poorly suited to the problem. Architecture is unclear.	No coherent AI solution design or architecture is presented.

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TECHNICAL IMPLEMENTATION & AI INTEGRATION (30%)

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
AI Method Selection & Justification (10%)	AI methods (e.g., Deep Learning models, LLMs, Fine-tuning) are perfectly chosen, expertly applied, and thoroughly justified for the problem.	AI methods are well-chosen and appropriately applied, with clear justification.	AI methods are generally suitable, but justification may be weak or alternatives not fully explored.	AI methods are questionable or ill-suited for the problem; justification is lacking.	Inappropriate or no AI methods are used; no justification.
Code Quality & Functionality (10%)	Code is exceptionally clean, well-documented, efficient, and robust. All functionalities are fully implemented and bug-free.	Code is clean, well-documented, and functional, with minor issues or inefficiencies.	Code is generally functional, but may have some bugs, poor documentation, or inefficiencies.	Code is buggy, poorly documented, or only partially functional.	Code is non-functional, unreadable, or missing significant components.
System Cohesion & Scalability (10%)	System components are seamlessly integrated, demonstrating high cohesion and strong consideration for scalability and modularity.	System components are well-integrated and cohesive, with reasonable attention to scalability.	Components are integrated, but cohesion may be weak, or scalability considerations are limited.	Components are loosely integrated or show significant cohesion issues; scalability is not addressed.	System lacks integration or is fundamentally incoherent.

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EVALUATION & REFINEMENT (20%)

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Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
Evaluation Methodology & Metrics (10%)	Evaluation methodology is rigorous, relevant metrics are perfectly chosen, and experiments are well-designed and executed.	Evaluation methodology is sound, relevant metrics are used, and experiments are well-conducted.	Evaluation methodology is acceptable, but may miss some relevant metrics or have minor flaws in experimental design.	Evaluation methodology is inadequate, uses inappropriate metrics, or lacks proper experimental design.	No meaningful evaluation or metrics are presented.
Performance Analysis & Iteration (10%)	Comprehensive and insightful analysis of performance, clearly identifying strengths, weaknesses, and demonstrating impactful iterative refinement.	Good analysis of performance with identification of strengths and weaknesses, and evidence of some iterative refinement.	Basic analysis of performance; some strengths/weaknesses noted, but limited evidence of refinement.	Superficial analysis of performance; little or no attempt at refinement.	No analysis of performance or demonstration of refinement.

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COMMUNICATION & DOCUMENTATION (15%)

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
Technical Report Quality (10%)	Report is professional, comprehensive, logically structured, and clearly communicates all aspects of the project. Exemplary writing style.	Report is well-structured and communicates project aspects clearly and professionally.	Report is generally clear but may lack some detail, organization, or professional polish.	Report is confusing, incomplete, poorly organized, or contains significant errors.	Report is unreadable or absent.
Presentation Effectiveness (5%)	Presentation is highly engaging, clear, concise, well-structured, and includes effective visuals/demonstrations. Strong defense of the project.	Presentation is clear, well-structured, and includes appropriate visuals/demonstrations. Good defense.	Presentation is generally clear but may lack engagement or polish; defense is adequate.	Presentation is disorganized or unclear; visuals are poor; defense is weak.	No presentation or extremely poor delivery.

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ETHICAL, LEGAL, & SOCIAL CONSIDERATIONS (10%)

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)
Consideration & Analysis (10%)	Thorough identification and insightful analysis of all relevant ethical, legal, and social implications. Proactive discussion of mitigation strategies.	Identified and discussed most relevant ethical, legal, and social implications with good analysis.	Identified some ethical, legal, or social implications, but analysis may be superficial or incomplete.	Limited identification or analysis of ethical, legal, or social considerations.	No recognition or discussion of ethical, legal, or social implications.

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CAPSTONE PROJECT TEAM RULES

- **Team Size:** Maximum of **three** members.
- **Active Participation:** All team members are expected to contribute equally to all project phases, from ideation to final presentation.
- **Regular Communication:** Teams must establish and maintain consistent communication channels
- **Documentation & Version Control:** All project work, including code and reports, must be systematically documented and managed using version control systems (e.g., Git).
- **Problem-Solving:** Teams are encouraged to collaborate on problem-solving, leveraging each member's strengths.
- **Ethical Conduct:** Adherence to ethical AI principles and academic integrity is mandatory throughout the project.
- **Deadline Adherence:** All interim and weekly and final deadlines must be met. Timely submission of deliverables is crucial.

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